

Research Article

Biomass and Carbon of *Eucalyptus* spp. Genotypes Estimated With UAV-LIDAR Using Area-Based Approach and Machine Learning

Carla Talita Pertille ¹, Alexandre Behling ¹, Carlos Roberto Sanquetta ¹,
Otávio Camargo Campoe ², Gueric Le Maire ³, Joannès Guillemot ³,
Geovanni Malatesta Barros ⁴, Emerson Roberto Schoeninger,⁴
and Ana Paula Dalla Corte ¹

¹Department of Forest Engineering, Federal University of Paraná-UFPR, Curitiba, Paraná, Brazil

²Department of Forest Sciences, Federal University of Lavras-UFLA, Lavras, Minas Gerais, Brazil

³Le Centre de Coopération Internationale en Recherche Agronomique Pour le Développement-CIRAD, Montpellier, Occitanie, France

⁴Bracell, Lençóis Paulista, São Paulo, Brazil

Correspondence should be addressed to Carla Talita Pertille; carlatapertille@gmail.com

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This study aimed to quantify aboveground biomass and aboveground carbon in the total, trunk, and canopy compartments of *Eucalyptus* spp. using UAV-LiDAR data, using modeling techniques. The study area is part of the EUCFLUX project located in São Paulo, Brazil, and consists of five experimental blocks with 25 *Eucalyptus* spp. genotypes. Forest inventory and destructive sampling to assess biomass took place in May 2023. UAV-LiDAR data were collected with a Zenmuse L1 sensor on a DJI Matrice 300 aircraft, and predictive models were developed based on the metrics selected through principal component analysis. RF obtained an adjusted R^2 of 0.952 and 0.959 for total and trunk aboveground biomass, respectively, and an RMSE of 5.42% for total aboveground biomass and 4.59% for trunk biomass. The adjusted R^2 values for total aboveground carbon and trunk carbon were 0.947 and 0.959, with RMSE of 5.38% and 4.54%, respectively. The XGB algorithm outperformed the other methods in the canopy compartment, with adjusted R^2 of 0.765 and 0.707 and RMSE of 13.49% and 15.06%, respectively. This study highlights the potential of UAV-LiDAR and nonparametric algorithms for accurately estimating carbon stock in *Eucalyptus* spp., contributing to climate change mitigation strategies.

Keywords: climate change; ecosystem services; nonparametric modeling; UAV-LiDAR metrics

1. Introduction

Brazil has one of the largest planted forest covers in the Earth with 10.2 million hectares [1], which equates to 12% of the total [2]. *Eucalyptus* spp. stands out as the most planted among the main types of trees planted in the country, with approximately 7.83 million hectares, representing 76% of the area of planted forests in the country [1]. Given this vast number, the leading role of forests in mitigating climate change stands out, especially through the conversion of

atmospheric carbon into forest biomass (a process carried out by forests) [3], in addition to their importance in the global carbon cycle [4].

Carbon stock in forest areas is mainly concentrated in the forest biomass, divided into aboveground biomass (AGB), belowground biomass (BGB), necromass (trunks, stumps, and dead branches), litter, and stable organic matter distributed in the soil profile [5]. The largest concentration of forest biomass is in the AGB [6], especially in tropical forests, and can be obtained by direct methods (forest

inventory and destructive sampling of individuals), and/or estimated indirectly (from allometric equations and data derived from remote sensing technologies) [7].

Given the spatial limitations of direct methods, especially due to the cost and execution time (since the tree is felled, sectioned and the components are weighed in situ) [8], nondestructive, continuous, and spatial approaches have stood out in AGB estimates, especially due to the application of remote sensing data, such as light detection and ranging (LiDAR) [9].

LiDAR is an active laser scanning sensor (light amplification by stimulated emission of radiation). Its operation is based on emitting laser beams at a specific wavelength with a high repetition frequency rate that penetrates forests, thus obtaining three-dimensional information with high resolution [10]. According to [11], these systems can be divided according to the pulse signal forms (discrete or continuous), the type of platform (aerial, terrestrial, or orbital), and the footprint size (small (~1 m), medium (~10–30 m), or large (~50 m)).

According to these sensor characteristics, two approaches can be used to estimate AGB using LiDAR technology: (i) individual-based approach or individual tree detection (ITD) and (ii) area-based approach (ABA). In this last method, attributes at the plot or stand level are extracted and applied in creating predictive AGB models [12].

This ABA enables calculating metrics which characterize the point cloud of an area of interest, which can be used in different predictive statistical models [13], such as traditional techniques (i.e., multiple linear regression) and machine learning (ML) algorithms. Estimating AGB in forest plantations using multiple linear regression has limitations, such as the difficulty in treating complex and nonlinear relationships or multicollinearity problems [14]. This fact motivated the search for ML techniques for predicting biomass and carbon stock, such as artificial neural networks (ANNs) [15], k-nearest neighbor (kNN) [16], random forest (RF) [14], support vector regression (SVR) [17], and extreme gradient boosting (XGB) [18].

In this context, robust estimates of carbon stock stored in the biomass of *Eucalyptus* spp. forests are becoming increasingly necessary and urgent, especially in addressing the climate emergency. In turn, this study is justified by estimating the biomass and carbon stock in multilayers (i.e., in different compartments) by combining a high-precision data source (unmanned aerial vehicle (UAV) LiDAR or UAV-LiDAR platform, with high point density) and ML techniques (such as ANN, kNN, RF, SVR, and XGB).

Although there are many studies using UAV-LiDAR to estimate AGB in *Eucalyptus* spp. plantations, there are still some limitations to consider. Most of these studies present insufficient field validation or require extensive field sampling to obtain a larger number of samples [19]. Another challenge concerns the need for allometric models calibrated according to species, age, and site conditions [19, 20]. Furthermore, the lack of reliable models to predict biomass by compartment (total, trunk, and canopy) hinders accurate estimation of carbon stocks in different parts of the tree. Other constraints involve variation in point cloud density

and quality, effects of flight conditions on data acquisition, low spatial and temporal transferability of models, and the cost and complexity of processing large volumes of UAV-LiDAR data [21, 22]. These gaps highlight the importance of this study, as improving more accurate and detailed methods helps optimize forest carbon monitoring, support climate change mitigation policies, and provide essential data for the sustainable management of *Eucalyptus* spp. plantations. Along these lines, this work aimed to estimate the AGB and aboveground carbon stock in different compartments (total, trunk, and canopy) focusing on unit area and with different modeling methods in *Eucalyptus* spp. genotypes using high-density UAV-LiDAR data.

2. Materials and Methods

2.1. Area Description. The area evaluated is located in the municipality of Itatinga, in the state of São Paulo, Brazil. The farm belongs to a forestry company in the pulp and paper industry and has an experimental area of 192.54 ha, in which this study was conducted (Figure 1). This area is linked to the Cooperative Program on Productivity and Carbon and Water Fluxes in *Eucalyptus* project—Phase 2 (*Programa Cooperativo sobre Produtividade e Fluxos de Carbono e Água em Eucalyptus—EUCFLUX*) developed in partnership with the Forestry Research Institute (Instituto de Pesquisas Florestais—IPEF) [23] and CIRAD (*Le Centre de coopération Internationale en Recherche Agronomique pour le Développement*).

The region's climate is classified as Cfa according to the Köppen climate classification system, meaning a humid and hot subtropical climate with hot summers and dry winters [24]. It is located at an altitude of 845 m, with an average annual temperature of 19.4°C and an average annual precipitation of 1319 mm [20]. The relief is mostly flat, with soils classified as typical dystrophic red latosol (*Latossolo Vermelho Distrófico Típico—LVd4*) and typical dystrophic red-yellow latosol (*Latossolo Vermelho-Amarelo Distrófico Típico—LVAd4*), with a medium sandy texture and moderate A horizon, and typical dystrophic red latosol (*Latossolo Vermelho Distrófico Típico—LVd2*), with a clayey texture and moderate A horizon [25].

The experimental area comprises a plantation of 5 blocks containing 25 genotypes of the genus *Eucalyptus* spp. planted at the end of 2018 with spacing of 3 × 2 m, totaling 6 m² per tree and 1666 trees.ha⁻¹. The high-trunk management system was implemented. Planting with a manual planter was performed after the harvest of the first rotation (8.5 years) in 2018, and replanting was carried out 30 days later. Planting management involved base and top-dressing fertilization over time, in addition to weed control through chemical weeding and leaf-cutter ant control with ant baits at different times of the cycle according to the climatic conditions and the plant development stage [26].

Each block contains approximately 2.80 ha and contains 25 plots representing the 25 genotypes planted. The total area of the block is 180 × 160 m, containing 25 plots of 36 × 32 m (12 rows × 16 trees), totaling 192 trees per plot. However, the useful area or the effective internal measurement area is composed of 30 × 20 m, totaling 600 m² or 0.06 ha,

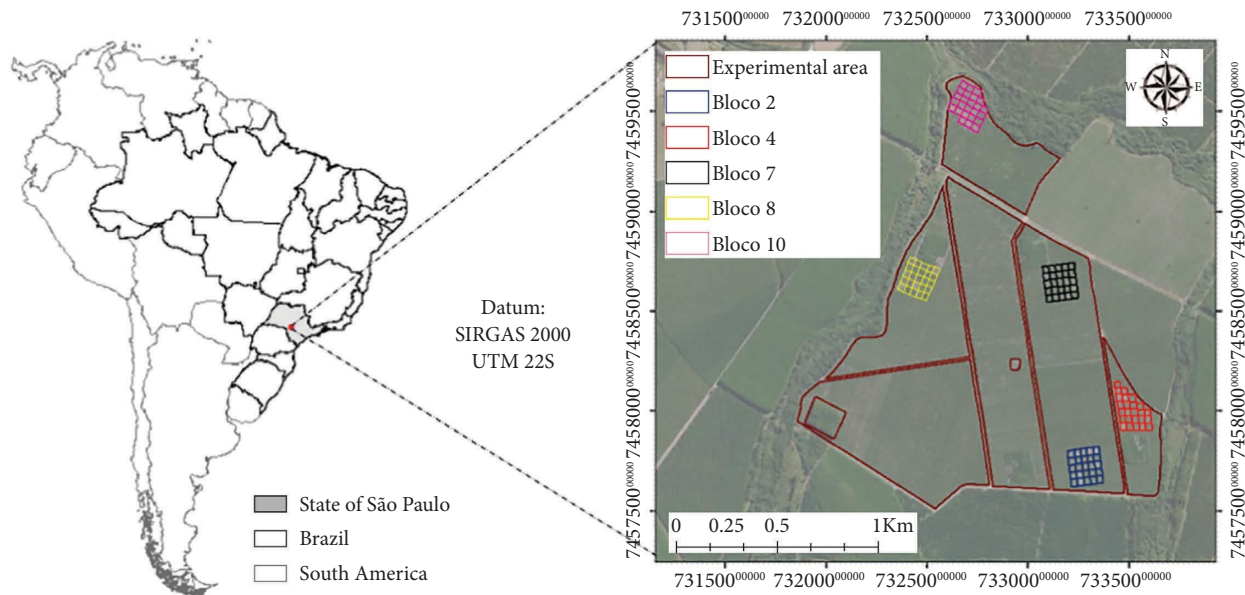


FIGURE 1: Experimental area with *Eucalyptus* spp. in Itatinga, São Paulo, Brazil.

containing 10 rows $\times$ 10 trees, meaning 100 trees per plot for each of the blocks. Of the 125 plots in the experimental area, only 124 plots were used in this study, since genotype 23 planted in block 10 did not present field measurements due to the high mortality rate after planting.

2.2. Forest Inventory. The forest inventory was conducted in the existing plots in the experiment in May 2023. The diameter at breast height (DBH) was measured in each tree with a caliper, and the total height of the trees located in the fourth and fifth rows was obtained using a Haglof Clinometer. The height of the remaining trees in each plot was subsequently determined from the Trorey hypsometric model (equation (1)), with an adjusted coefficient of determination (adjusted R^2) of 0.724, standard error of the estimate (Syx) of 2.36 m, and standard error of the percentage estimate (Syx (%)) of 11.29%:

$$TH = (-4.7496 + 2.9752 * DBH + -0.0728 * DBH^2), \quad (1)$$

where TH is total height (m); DBH is diameter at breast height (cm).

Next, 150 trees (6 trees per genotype) not belonging to the plots were selected by diameter class using the standard deviation method. Tree volume was estimated using Smalian's method to obtain the plot volume, with measurements taken at 0.10, 0.30, 1, and every 3 m up to the tree top. The Schumacher–Hall model (equation (2)) was adopted for the volumetric quantification of the genotypes whose fit presented an adjusted coefficient of determination (adjusted R^2) of 0.985 and a standard error of the estimate in percentage (Syx (%)) of 6.47%:

$$\ln = -10.2992 + 1.7642 * \ln + 1.2262 * \ln, \quad (2)$$

where $\ln$ is natural logarithm; v is total volume (m^3); TH is total height (m); and DBH is diameter at breast height (cm).

2.3. Quantification of AGB and Aboveground Carbon. The AGB in the field was determined considering the above-ground tree components (wood, bark, live branches, dead branches, and leaves) per tree unit ($kg \cdot tree^{-1}$) using the 150 trees measured during the forest inventory.

After the trees were felled, the components were separated and weighed to determine the wet weight. Thus, 2.5-cm-thick discs were removed from the positions corresponding to the base (10 cm), at DBH (1.30 m), and every 3 m of height of the felled tree to collect the wood samples.

The samples were transported to the Forestry and Forest Hydrology Laboratory, located at the Itatinga Experimental Station of Forest Sciences, São Paulo, Brazil. The samples were initially subjected to the drying process at $65^\circ C$ to obtain the dry weight, requiring 4 days for leaves, dead branches and bark, 7 days for live branches, and 3 weeks for wood. After obtaining the respective dry weights, the total dry matter percentage (%) (equation (3)) of each evaluated sample was calculated, and then, the total dry weight or total dry biomass (equation (4)) of each tree component is as follows:

$$DM_i = \frac{DW_i}{WW_i}, \quad (3)$$

$$TDM_i = DM_i * TWW_i, \quad (4)$$

where DM_i is dry matter of the i th tree component (%), DM_i is dry weight of the sample of the i th tree component (g), WW_i is wet weight of the sample of the i th tree component (g), TDM_i is total dry mass of the i th tree component (kg), and TWW_i is total wet weight of the i th tree component (kg).

Then, the average carbon content of 46.08% derived from the study of [27] in *Eucalyptus* spp. aged 4.7 years was adopted to obtain the aboveground carbon stock. The same value of carbon content was applied to all compartments, since [27] analyzed multiple *Eucalyptus* species—consistent with our dataset comprising 25 genotypes of *Eucalyptus* spp.—and the tree age was close to that of the present study (~4 years). The dry biomass of each compartment analyzed was multiplied by the average carbon content, resulting in the carbon stock per component. Thus, the sum of the carbon stock per tree component resulted in the total aboveground carbon stock (kg.tree^{-1}).

In turn, the Schumacher–Hall model was used with the inclusion of the genotypes as dummy variables to quantify the AGB per compartment (total, trunk, and canopy) per genotype (equations (5)–(7)). The sum of the AGB per compartment of all trees belonging to each genotype was subsequently divided by 1000 and multiplied by the proportionality factor of 16.67 ($10,000/600 \text{ m}^2$), resulting in the AGB of each compartment in $\text{t}\cdot\text{ha}^{-1}$:

$$\ln tb = \beta_0 + \beta_1 * \ln DBH + \beta_2 * \ln HT + \text{genotype} + \varepsilon, \quad (5)$$

$$\ln trb = \beta_0 + \beta_1 * \ln DBH + \beta_2 * \ln HT + \text{genotype} + \varepsilon, \quad (6)$$

$$\ln tcb = \beta_0 + \beta_1 * \ln DBH + \beta_2 * \ln HT + \text{genotype} + \varepsilon, \quad (7)$$

where tb is total AGB (kg), trb is AGB of trunk (kg), tcb is AGB of canopy (kg), β_i is model coefficient, TH is total height (m), DBH is diameter at breast height (cm), and ε is the error associated with the model.

Finally, the AGB ($\text{t}\cdot\text{ha}^{-1}$) of each compartment per genotype was multiplied by the average carbon content [23], resulting in the aboveground carbon stock per compartment (total, trunk, and canopy) per tree.

2.4. UAV-LiDAR Scanning. The UAV-LiDAR data were acquired using a DJI Matrice 300 UAV [24] and a Zenmuse L1 sensor [28] in May 2023. The flight altitude was 80 m, with a speed of $9 \text{ m}\cdot\text{s}^{-1}$ and a coverage swath width of 450 m. The Zenmuse L1 sensor can record up to 3 returns, collecting data with a wavelength of 905 nm, a scan angle of 45° , a field of view of 70.4° (horizontal) and 77.2° (vertical), and a vertical accuracy of 5 cm at 50 m and a horizontal accuracy of 10 cm at 50 m [28], resulting in an average point density of $389.30 \text{ points}\cdot\text{m}^{-2}$.

The point cloud derived from the UAV-LiDAR survey was processed in R software, Version 4.2.0 [29] with the aid of the *lidR* library [13]. The ground returns were initially classified to only represent the ground [26], for which the cloth simulation filter (CSF) was used. After classifying the

points, the returns extracted as ground were used to generate the digital terrain model (DTM) using the kNN algorithm with inverse distance weighting (IDW) [30]. The digital surface model (DSM) was obtained by the p2r (point to raster) algorithm, which assigns the elevation of the highest point to each pixel [13].

Next, the point cloud was normalized using the *normalize_height* function [13]. After normalization, the canopy height model (CHM) was extracted using the p2r (point to raster) method [13, 30]. All digital models (DTM, DSM, and CHM) were extracted with 0.5 m resolution. The normalized point cloud and the CHM were subsequently extracted considering the parcel boundaries of each sub-block. The metrics related to height (maximum, mean, standard deviation, skewness, kurtosis, and entropy) in addition to the height percentiles and the cumulative return percentages were extracted using the *lidR* library [13] in R Version 1.0. 4.2.0 software [29].

2.5. Modeling of AGB and Aboveground Carbon. First, the UAV-LiDAR metrics were selected from principal component analysis (PCA) to build the models using multiple linear regression. After selecting the metrics, the dataset was divided, considering 70% (88 data) for the formation of a training set and 30% (36 data) for a validation set using the “*caret*” package [31] in the R Version 4.2.0 software [29].

Then, the 5 candidate metrics for building the models were selected by the stepwise method, adding or removing variables by the “*lmStepAIC*” method in the “*caret*” package [31]. The models derived from the multiple linear regression technique were evaluated for residual normality (Shapiro–Wilk test) [32], independence of the residuals (Durbin–Watson test), and homogeneity of variances (Breusch–Pagan test) [10]. All analyses were performed in the R environment [29] at 95% probability. In addition, traditional fitting parameters were also considered, such as adjusted coefficient of determination (adjusted R^2), absolute ($\text{Syx t}\cdot\text{ha}^{-1}$) and percentage ($\text{Syx } \%$) standard error of the estimate, and absolute ($\text{t}\cdot\text{ha}^{-1}$) and percentage (%) root mean square error (RMSE).

The parameters of the ML algorithms tested are shown in Table 1. All algorithms were tested with the *expand.grid* function of the *caret* package [31] to create a grid of hyperparameter combinations to be tested. The ANNs were applied using the *nnet* method (*nnet* package) [33]. The other algorithms were tested in the *caret* package [31] using the *knn*, *rf*, *svmLinear*, and *xgbTree* methods. The model fit and parameter optimization steps described above were performed in the R software [29].

The following agreement measures were considered to validate and select the best predictive model for AGB and aboveground carbon: coefficient of agreement (A) (equation (8)), measure of agreement (p) (equation (9)); improved concordance correlation coefficient (CC2) (equation (10)), and agreement parameter (equation (11)) [34–38]:

TABLE 1: Machine learning algorithms tested for predicting aboveground biomass and aboveground carbon in different compartments of *Eucalyptus* spp. genotypes in São Paulo, Brazil.

Algorithm	Parameter	Variation	Description
ANN	Size	1 to 10	Number of neurons in the hidden layer
ANN	Decay	0.1 to 1	Regularization term (weight penalty)
kNN	k	10, 12, and 14	Number of neighbors
RF	mtry	10, 12, 14, and 16	Number of randomly selected predictor variables
SVR	Sigma	0.05 to 0.1	Radial kernel function parameter
SVR	C	5, 7, and 10	Regularization parameter
XGB	max_depth	3, 5, and 7	Maximum tree depth
XGB	eta	0.01, 0.05, and 0.1	Learning rate
XGB	Gamma	0, 0.01 and 0.02	Penalty for node splitting
XGB	colsample_bytree	0.2, 0.4, and 0.6	Percentage of columns sampled per tree
XGB	min_child_weight	1, 3, and 5	Minimum number of samples for node splitting
XGB	Subsample	0.7, 0.8, and 0.9	Percentage of data used in each tree

Abbreviations: ANN, artificial neural network; kNN, k-nearest neighbor; RF, random forest; SVR, support vector regression; XGB, extreme gradient boosting.

$$A = 1 - \frac{\sum (y_i - k_i)^2 + \sum (x_i - k_i)^2}{\sum (y_i - k)^2 + \sum (x_i - k)^2}, \quad (8)$$

$$p = 1 - \frac{\text{MSE}}{(1/(n^2)) \sum_{i=1}^n (y_i - x_i)^2}, \quad (9)$$

$$\text{CC2} = r * A_p, \quad (10)$$

$$\lambda = 1 - \frac{\sum (y_i - x_i)^2}{\sum (y - y_i)^2 + \sum (x_i - \bar{x})^2 + n(\bar{y} - \bar{x})^2} + k, \quad (11)$$

where A : coefficient of agreement; y_i : AGB estimated and aboveground carbon estimated in the different compartments observed in the field ($\text{t}\cdot\text{ha}^{-1}$); x_i : AGB observed and aboveground carbon observed in the different compartments ($\text{t}\cdot\text{ha}^{-1}$); k_i : mean of y_i and x_i for tree i ; k : mean of all y_i and x_i values; p : measure of agreement; n : number of observations; MSE: mean square error; CC2: improved concordance correlation coefficient; r : correlation coefficient; A_p : modified measure of accuracy; λ : agreement parameter; $\bar{y}$: mean of y ; and $\bar{x}$: mean of x .

Estimates of AGB and aboveground carbon per compartment were analyzed using the Bland–Altman plot [39] in R software [29], with the blandr package [40].

The complete sequence of methodological steps described is summarized in the flowchart in Figure 2.

3. Results

3.1. Destructive Sampling: Total AGB and Total Aboveground Carbon. With the hypsometric relationship adjusted by the Trorey model and the volumetric quantification by the Schumacher–Hall model, the biometric variables of the evaluated genotypes (Table 2) presented an average diameter of 12.86 cm and an average height of 20.63 m. Block 7 presented the tallest trees (31.10 m), while the thickest trees were predominant in block 10 (13.35 cm). The largest

volumetric stock occurred in block 10 ($282.21 \text{ m}^3\cdot\text{ha}^{-1}$). In turn, block 2 had the smallest volume ($236.65 \text{ m}^3\cdot\text{ha}^{-1}$). It is worth mentioning that there was no data collection in block 10 for genotype 23 due to its high mortality rate.

Regarding the components evaluated in the scaled trees, the highest biomass concentration on average was noted in the wood, with $64.20 \text{ kg}\cdot\text{tree}^{-1}$ (78.96%), followed by the bark with $8.17 \text{ kg}\cdot\text{tree}^{-1}$ (10.05%), live branches with $4.74 \text{ kg}\cdot\text{tree}^{-1}$ (5.83%), leaves with $2.80 \text{ kg}\cdot\text{tree}^{-1}$ (3.45%), and dead branches with $1.39 \text{ kg}\cdot\text{tree}^{-1}$ (1.71%) (Figure 3).

Fitting the regression model considering the genotypes as dummy variables for estimating AGB in different compartments (Table 3) obtained coefficients of determination higher than 0.95 and errors of approximately 10%, except for the canopy compartment (branches, dead branches, and leaves), whose fit had an adjusted R^2 of 0.8076 and errors close to 30%.

Table 4 summarizes the determination of AGB in different compartments by the Schumacher–Hall model and the estimate of the aboveground carbon stock present in the *Eucalyptus* spp. genotypes. The mean total AGB was $111.22 \text{ t}\cdot\text{ha}^{-1}$, and the total aboveground carbon stock had an average of $51.25 \text{ t}\cdot\text{ha}^{-1}$. The trunk compartment had, on average, an AGB of $100.01 \text{ t}\cdot\text{ha}^{-1}$ and an aboveground carbon of $46.09 \text{ t}\cdot\text{ha}^{-1}$. The canopy compartment had a mean AGB of $10.79 \text{ t}\cdot\text{ha}^{-1}$ and aboveground carbon of $4.97 \text{ t}\cdot\text{ha}^{-1}$. It is noted that genotype 21 presents the highest biomass and carbon concentrations in all compartments. In contrast, genotype 25 has the lowest amounts of biomass and carbon.

3.2. Modeling of AGB and Aboveground Carbon by Compartment. First, the 36 extracted UAV–LiDAR metrics were subjected to an exploratory analysis to identify outliers and select the variables using PCA. Thus, 8 metrics that presented values equal to 0 were excluded. PCA was subsequently applied, which identified that the first two principal components can explain 81% of the data variation. Next, the remaining 28 metrics were analyzed again and subjected to a correlation analysis to identify multicollinear metrics, or highly correlated. In turn, 20 metrics related to the mean height, percentiles of the height

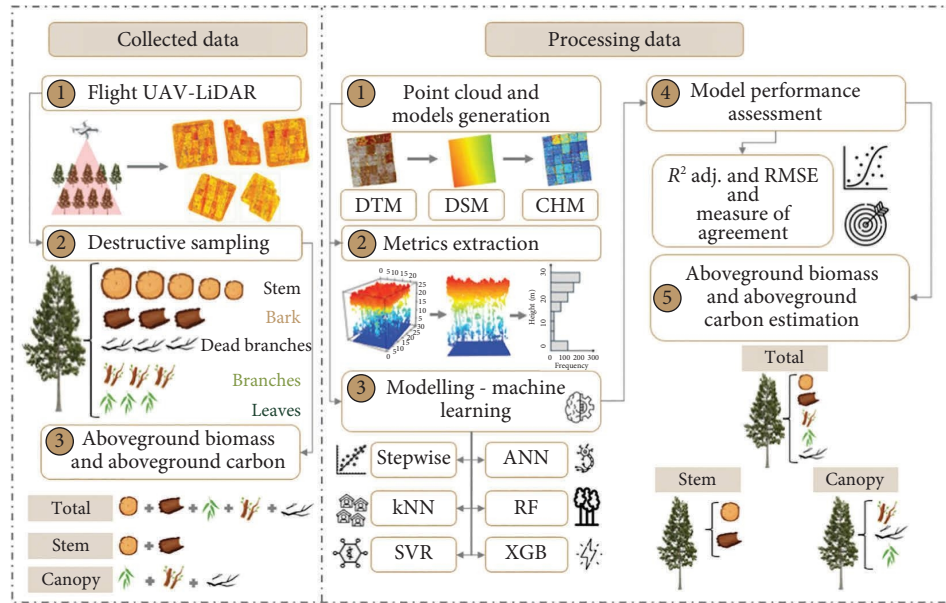


FIGURE 2: Flowchart of UAV-LiDAR data processing for estimating aboveground biomass and aboveground carbon at the compartment level in *Eucalyptus* spp. in Itatinga, São Paulo, Brazil.

TABLE 2: Biometric variables of *Eucalyptus* spp. genotypes in an experimental area in Itatinga, São Paulo, Brazil.

Block	N.trees.ha	md (cm)	G ($\text{m}^2 \cdot \text{ha}^{-1}$)	maxh (m)	mh (m)	V ($\text{m}^3 \cdot \text{ha}^{-1}$)
2	1398	12.71	23.27	26.60	20.37	236.65
4	1481	12.72	24.64	28.20	20.56	252.07
7	1538	12.69	25.45	31.10	20.80	265.15
8	1544	12.84	26.21	27.80	20.72	270.61
10	1436	13.35	27.02	30.20	20.70	282.21
Mean	1480	12.86	25.32	31.10	20.63	261.34

Note: Em que: N.trees: number of trees measured per hectare; md: mean diameter (cm); G: basal area ($\text{m}^2 \cdot \text{ha}^{-1}$); maxh: maximum total height (m); mh: mean total height (m); V: average total volume ($\text{m}^3 \cdot \text{ha}^{-1}$). The bold values refer to the mean of each variable.

distribution located between 30% and 65%, cumulative percentages of returns between 10% and 90% of the height, and the percentage of returns above the average were excluded. Of the remaining 8 metrics, the 3 metrics with the lowest weight in the PCA were excluded, namely, mean height, standard deviation of height, and 95% height percentile. As a result, 5 metrics were used in creating the predictive models of AGB and aboveground carbon in the different compartments using ML and regression models, including height percentile at 70%, 75%, 80%, 85%, and 90%, whose model performed well in cross-validation (R^2 of 0.73 and an RMSE of 3.05).

Fitting of the models constructed by stepwise and ML for predicting AGB and aboveground carbon in the different compartments is summarized in Table 5. The models fitted by the stepwise method demonstrated an explanatory power (adjusted R^2) which varied between 61% and 78% for the response variables analyzed. The RMSE percentage varied from 10.18% to 18.73%, with the highest value observed in the canopy compartment.

As a result, the RF and XGB techniques generally stood out among the tested ML methods, while ANN remained in third place. RF presented adjusted R^2 values above 0.95 for

predicting AGB and aboveground carbon in the total compartment, indicating excellent predictive capacity. Next, XGB stood out with an adjusted R^2 of 0.893 for total AGB and total aboveground carbon.

The RF algorithm again had the best fit in estimating biomass and carbon located in the trunk with an adjusted R^2 of 0.959 and 0.961, respectively, and errors below 5%, surpassing the results obtained with XGB and ANN. However, the XGB algorithm presented the best fit statistics in the canopy compartment for both biomass and carbon, with an adjusted R^2 of 0.844 for biomass and 0.924 for carbon. RF also presented an equally satisfactory adjustment, but slightly lower than XGB, with adjusted R^2 of 0.765 for biomass and 0.707 for carbon. In contrast, the models originating from the kNN and SVR techniques presented an adjusted R^2 lower than 0.782 for all compartments, with the lowest value observed in predicting canopy biomass, which was 0.285. These methods also presented the highest prediction errors (between 9.94% and 27.06%), especially in the canopy compartment.

The models derived from multiple linear regression for predicting total aboveground and trunk biomass, as well as total aboveground and trunk carbon, presented residuals

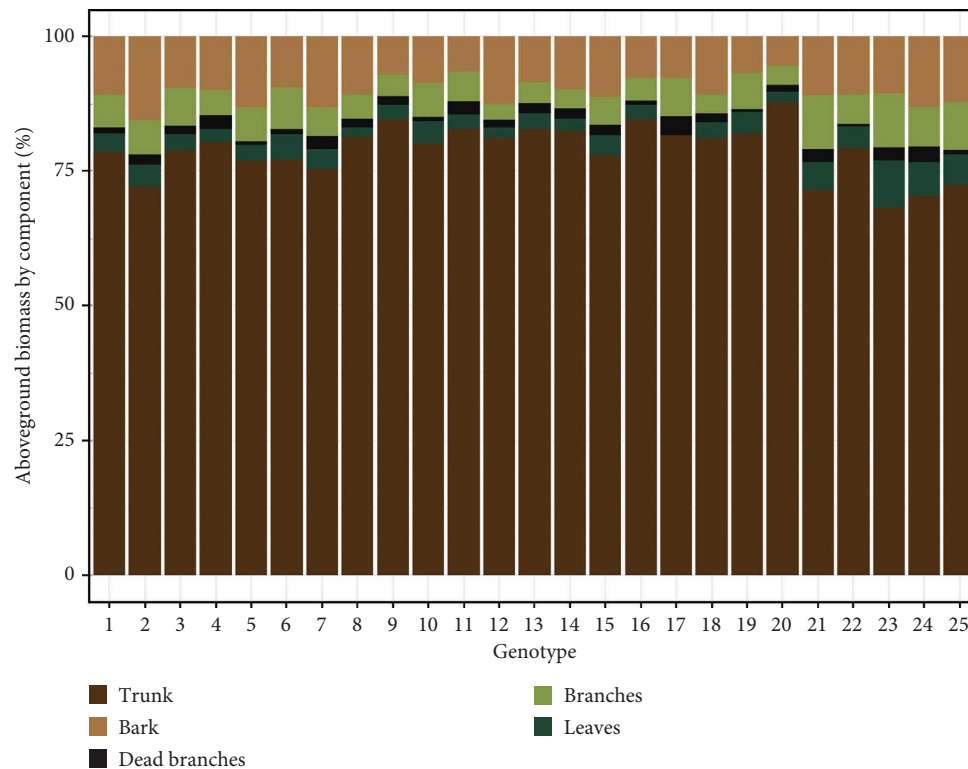


FIGURE 3: Percentage of total aboveground biomass per compartment in *Eucalyptus* spp. trees in Itatinga, São Paulo, Brazil.

TABLE 3: Estimation of aboveground biomass in different compartments (total, trunk, and canopy) by the Schumacher–Hall model in *Eucalyptus* spp. in Itatinga, São Paulo, Brazil.

Compartment	R^2 adj	Syx (kg.tree ⁻¹)	Syx (%)	RMSE (kg.tree ⁻¹)	RMSE (%)
Total	0.960	8.65	10.64	7.84	9.65
Canopy	0.808	3.01	33.72	2.61	29.23
Trunk	0.961	7.46	10.31	6.77	9.35

Note: R^2 adj: adjusted coefficient of determination; Syx: standard error of the estimate (kg.tree⁻¹); Syx (%): standard error of the percentage estimate (%); RMSE: root mean square error (kg.tree⁻¹); RMSE (%): root mean square error percentage (%).

with normal distribution, although these were independent in all compartments. Only the predictive models for biomass and carbon in the canopy compartment showed homoscedasticity of the variances.

Figure 4 illustrates the graphical distribution of the model residuals built with ML and multiple linear regression for the basis used in validating the models. The residual heteroscedasticity is noted in all the algorithms tested; however, to a lesser extent in the random forest model in the total aboveground and trunk compartments, and in the XGB model in the canopy compartment.

Table 6 contains the agreement measures evaluated for the models tested in the prediction of AGB and aboveground carbon in the compartments. Considering the total basis, fit basis, and validation basis, the RF algorithm presented the best agreement measures (closest to 1) in the total aboveground and trunk compartments. Next, there are XGB, ANN, SVR, and kNN. In turn, the XGB algorithm was the best for the canopy compartment, followed by RF, ANN, SVR, and kNN.

Figure 5 illustrates the Bland–Altman concordance plot for the predictive models of AGB and aboveground carbon of the total, trunk, and canopy compartments in *Eucalyptus* spp. considering the validation base.

Based on the fit and validation metrics, the agreement measures and the Bland–Altman plots, the RF algorithm was elected as the best for estimating total aboveground and trunk biomass, as well as for evaluating total aboveground carbon stock and trunk carbon, while the models built by XGB were more efficient in predicting biomass and carbon in the canopy compartment in *Eucalyptus* spp. genotypes from UAV-LiDAR metrics.

4. Discussion

This study aimed to quantify biomass and carbon stock considering the total, trunk, and canopy compartments of *Eucalyptus* spp. genotypes using the ABA approach based on integrating UAV-LiDAR data and different modeling techniques.

TABLE 4: Average aboveground biomass ($\text{t}\cdot\text{ha}^{-1}$) and average aboveground carbon ($\text{t}\cdot\text{ha}^{-1}$) per compartment in *Eucalyptus* spp. genotypes in São Paulo, Brazil.

Genotype	Aboveground biomass ($\text{t}\cdot\text{ha}^{-1}$)			Aboveground carbon ($\text{t}\cdot\text{ha}^{-1}$)		
	Total	Trunk	Canopy	Total	Trunk	Canopy
1	110.68	100.21	10.10	51.00	46.18	4.65
2	117.39	106.62	11.11	54.09	49.13	5.12
3	114.26	103.43	10.97	52.65	47.66	5.06
4	131.72	115.87	14.65	60.70	53.39	6.75
5	119.30	107.62	10.71	54.98	49.59	4.93
6	105.90	94.35	11.26	48.80	43.48	5.19
7	121.67	107.93	13.78	56.06	49.73	6.35
8	93.80	86.52	7.39	43.22	39.87	3.41
9	126.32	112.59	12.08	58.21	51.88	5.56
10	120.75	107.68	12.89	55.64	49.62	5.94
11	122.54	109.14	13.14	56.47	50.29	6.05
12	112.07	104.29	7.51	51.64	48.06	3.46
13	105.33	97.22	8.50	48.54	44.80	3.92
14	117.32	109.75	7.40	54.06	50.57	3.41
15	127.02	112.62	14.63	58.53	51.90	6.74
16	86.02	81.41	5.19	39.64	37.51	2.39
17	125.29	111.82	13.32	57.73	51.53	6.14
18	124.00	112.25	10.38	57.14	51.72	4.78
19	119.18	105.82	13.17	54.92	48.76	6.07
20	124.16	113.32	9.77	57.22	52.22	4.50
21	137.61	117.36	17.44	63.41	54.08	8.04
22	128.80	116.32	12.43	59.35	53.60	5.73
23	80.96	67.65	11.90	37.31	31.17	5.48
24	70.14	63.69	6.41	32.32	29.35	2.95
25	32.31	28.34	3.89	14.89	13.06	1.79
Mean	111.22	100.01	10.79	51.25	46.09	4.97
SD	25.40	22.57	3.46	11.70	10.40	1.60

Note: Mean: mean. The bold values refer to the mean of each variable. Abbreviation: SD, standard deviation.

According to the dendrometric variables (Table 2) from the field inventory, variations in the growth of genotypes between blocks are noted, evidencing the genotype–environment interaction and its impact on biomass production, which was greater in genotypes 20 and 18 and lower in genotypes 25 and 16 (Table 4). The trunk (wood and bark) represented 89% of the total AGB of the genotypes in the AGB distribution in the tree components. The greater biomass accumulation in the wood is due to the age of the genotypes, since there was a reduction in the canopy biomass and an increase in the biomass proportion in the wood and bark with the development of these plantations, in addition to genetic improvement which indirectly selects plants based on their biomass allocation in the trunk. Factors such as site conditions, species and genetic materials, climatic conditions, planting density, and rotation duration can also influence the biomass distribution in storage reservoirs [41].

The wood was also the compartment with the greatest contribution to the biomass of plantations aged between 4 and 6 years of *Eucalyptus urophylla* $\times$ *Eucalyptus grandis* located in Brasília and Paranoá, in the Federal District, Brazil. Wood accounted for between 69% and 86% of the biomass in the four analyzed areas, followed by the other

components: bark (6%–11%), roots (1%–10%), branches (1%–9%), and leaves (1%–4%) [42].

The AGB and carbon in different compartments (Table 3), as estimated by the Schumacher–Hall model with the inclusion of genotypes as dummy variables, were consistent with research conducted on similar genotypes under different conditions, according to [42]. In this study, 4-year-old *E. urophylla* $\times$ *E. grandis* plantations stored an average of $41.75 \text{ kg}\cdot\text{tree}^{-1}$ of carbon in the aboveground compartments [39]. The results in another study conducted in two experimental sites with contrasting climatic conditions belonging to the Cooperative Research Program for Tolerance of *Eucalyptus* spp. to Water, Thermal and Biotic Stresses (TECHS) of IPEF indicated that the *E. urophylla* and *E. grandis* $\times$ *E. camaldulensis* plantations stored on average $38.7 \text{ t}\cdot\text{ha}^{-1}$ of carbon at 3 years of age and $81.2 \text{ t}\cdot\text{ha}^{-1}$ of carbon at the end of the rotation (6 years) [43].

The results of this study regarding total AGB ($111.22 \text{ t}\cdot\text{ha}^{-1}$ at 4.5 years old) are in agreement with the literature. For example, [44] quantified the total AGB of five 4-year-old *Eucalyptus* spp. genotypes located in Eldorado do Sul in Rio Grande do Sul (Brazil), finding similar values for similar genotypes: *E. benthamii* ($92.19 \text{ t}\cdot\text{ha}^{-1}$), *E. benthamii* (area 2) ($102.04 \text{ t}\cdot\text{ha}^{-1}$), *E. saligna* ($89.25 \text{ t}\cdot\text{ha}^{-1}$), *E. dunni* ($68.40 \text{ t}\cdot\text{ha}^{-1}$), *E. urolobulus* ($117.52 \text{ t}\cdot\text{ha}^{-1}$), and *E. urograndis* ($101.80 \text{ t}\cdot\text{ha}^{-1}$).

The disparities found among the algorithms used in this study in biomass and carbon estimates resulting from the use of each algorithm are directly related to four main factors: (i) characteristics of the plantation and experimental area (spacing, age, and topography); (ii) UAV-LiDAR dataset (point density, interaction of laser returns with the canopy, and type of metrics used) and ABA approach; (iii) genotype–environment interactions (nutritional efficiency, sunlight absorption and photosynthetic capacity, resistance, and tolerance to adverse conditions); and (iv) optimization parameters of each algorithm.

The ABA approach is widely used in the forestry area, especially for inventory and mapping, but it is limited to the use of metrics derived from the vertical distribution of LiDAR returns [12]. Thus, the metrics contained in the cell size used comprise the main source of analysis for this approach [12].

In this regard, the influence of the plots should also be considered, since the number, size, shape, and accuracy of their geolocation interfere with the accuracy of biomass estimates [45]. Only 124 plots were used in this study—plots of 0.06 ha and rectangular shape. Reference [46] recommends that a greater number of plots and larger dimensions can reduce geolocation errors and consequently the biomass estimation process.

Genotype and environment interactions also influenced biomass production and carbon stock, since nutritional efficiency among these interactions can vary for each species [47] due to morphological characteristics and physiological characteristics (photosynthetic rate and the ability to maintain normal metabolism with low nutrient content in the tissues) [47].

TABLE 5: Fitting parameters of the predictive models of aboveground biomass and aboveground carbon ($\text{t}\cdot\text{ha}^{-1}$) in different compartments (total, trunk, and canopy) of *Eucalyptus* spp.

Compartments	Algorithm	Fitting (70% of the data)				Validation (30% of the data)					
		R^2 adj.	Syx ($\text{t}\cdot\text{ha}^{-1}$)	Syx (%)	RMSE ($\text{t}\cdot\text{ha}^{-1}$)	RMSE (%)	MAE	MPE	MSE	RMSE ($\text{t}\cdot\text{ha}^{-1}$)	RMSE (%)
Total aboveground biomass	Stepwise	0.744	13.30	12.04	13.14	11.90	9.08	8.64	134.69	11.61	10.25
	ANN	0.829	10.55	9.56	10.43	9.45	9.59	8.89	159.50	12.63	11.44
	kNN	0.758	12.54	11.36	12.40	11.23	10.51	11.09	174.47	13.21	11.96
	RF	0.952	6.05	5.48	5.99	5.42	9.12	8.48	133.64	11.56	10.47
	SVR	0.719	13.94	12.62	13.45	12.18	8.66	7.63	130.72	11.43	10.10
	XGB	0.893	8.33	7.54	8.23	7.46	9.06	8.22	137.47	11.72	10.36
Total aboveground carbon	Stepwise	0.744	6.13	12.04	6.06	11.90	4.18	8.64	28.60	5.35	10.25
	ANN	0.807	5.16	10.15	5.11	10.03	4.41	8.77	33.34	5.77	11.35
	kNN	0.758	5.78	11.36	5.71	11.23	4.84	11.09	37.05	6.09	11.96
	RF	0.952	2.77	5.44	2.74	5.38	4.21	8.47	28.15	5.31	10.17
	SVR	0.719	6.42	12.62	6.20	12.18	3.99	7.63	27.76	5.27	10.10
	XGB	0.893	3.84	7.56	3.80	7.47	4.20	8.28	29.43	5.43	10.66
Trunk biomass	Stepwise	0.780	10.39	10.42	10.15	10.18	8.97	10.48	136.20	11.67	11.59
	ANN	0.853	8.24	8.26	8.15	8.17	8.34	8.56	130.44	11.42	11.45
	kNN	0.782	10.03	10.06	9.92	9.94	10.09	13.91	162.93	12.76	12.80
	RF	0.959	4.63	4.64	4.57	4.59	8.18	8.90	116.14	10.78	10.81
	SVR	0.763	10.78	10.81	10.41	10.44	8.59	8.64	135.90	11.66	11.58
	XGB	0.927	5.82	5.83	5.75	5.77	8.14	8.46	121.42	11.02	10.94
Trunk carbon	Stepwise	0.780	4.79	10.42	4.68	10.18	4.13	10.48	28.92	5.38	11.59
	ANN	0.848	3.86	8.39	3.81	8.30	3.89	8.81	28.01	5.29	11.52
	kNN	0.782	4.62	10.06	4.57	9.94	4.65	13.91	34.60	5.88	12.80
	RF	0.961	2.11	4.59	2.09	4.54	3.81	9.16	25.29	5.03	10.84
	SVR	0.763	4.97	10.81	4.80	10.44	3.96	8.64	28.86	5.37	11.58
	XGB	0.932	2.58	5.61	2.55	5.54	3.73	8.37	26.31	5.13	11.16
Canopy biomass	Stepwise	0.610	2.23	20.71	2.02	18.73	2.24	25.55	9.31	3.05	28.22
	ANN	0.609	1.90	17.60	1.88	17.40	2.18	23.29	8.24	2.87	26.62
	kNN	0.564	2.29	21.26	2.27	21.02	2.58	27.78	9.54	3.09	28.65
	RF	0.844	1.54	14.29	1.52	14.13	2.64	28.34	8.89	2.98	27.66
	SVR	0.285	3.02	28.04	2.92	27.06	2.22	23.56	6.45	2.54	23.49
	XGB	0.765	1.47	13.64	1.45	13.49	1.81	18.65	5.39	2.32	21.46
Canopy carbon	Stepwise	0.610	1.03	20.71	0.93	18.73	1.03	25.53	1.98	1.41	28.22
	ANN	0.609	0.87	17.61	0.86	17.41	1.00	23.16	1.66	1.29	25.97
	kNN	0.557	0.93	18.73	0.92	18.52	0.99	23.58	1.48	1.22	24.48
	RF	0.924	0.55	11.17	0.55	11.04	0.91	21.16	1.21	1.10	22.07
	SVR	0.416	1.26	25.34	1.07	21.44	1.01	23.77	1.54	1.24	24.92
	XGB	0.707	0.76	15.24	0.75	15.06	0.88	20.31	1.17	1.08	21.79

Note: R^2 adj.: adjusted coefficient of determination; Syx: standard error of the estimate ($\text{t}\cdot\text{ha}^{-1}$); Syx (%): standard error of the percentage estimate (%); RMSE: root mean square error ($\text{t}\cdot\text{ha}^{-1}$); RMSE (%): root mean square error percentage (%); MAE: mean absolute error ($\text{t}\cdot\text{ha}^{-1}$); MPE: mean percentage error; MSE: mean squared error ($\text{t}\cdot\text{ha}^{-1}$). The bold values refer to the mean of each variable.

Abbreviations: ANN, artificial neural network; kNN, k-nearest neighbor; RF, random forest; SVR, support vector regression; XGB, extreme gradient boosting.

The modeling by biomass and carbon compartment revealed that in the total aboveground and trunk compartments, random forest was the best algorithm; in the canopy compartment, it was XGB. However, the application of the ML algorithms for estimating aerial biomass and aerial carbon in multicompartment (as in this research) of clonal plantations of *Eucalyptus* spp. in Brazil using UAV-LiDAR data is still in its infancy.

The RF algorithm combines predictions from multiple decision trees using bootstrap sampling, randomly selecting subsets of variables at each split. This reduces variance in complex predictions [14, 48], thereby mitigating the risk of overfitting and increasing accuracy [49]. [50] emphasize

that, for data obtained through remote sensing techniques, RF's high performance is due to its ability to handle high-dimensional data, such as variables derived from LiDAR data (mean height, height percentiles, point density, intensity metrics). However, its performance depends on hyperparameter settings (number of trees, mtry, minimum node size) and the validation method, since random validation can lead to overestimation in spatially correlated data. Therefore, spatial cross-validation is recommended [48].

RF has demonstrated consistency in estimating biomass from LiDAR data in forestry applications. For instance, [17] compared multiple algorithms and found that RF outperformed multiple linear regression in predicting forest

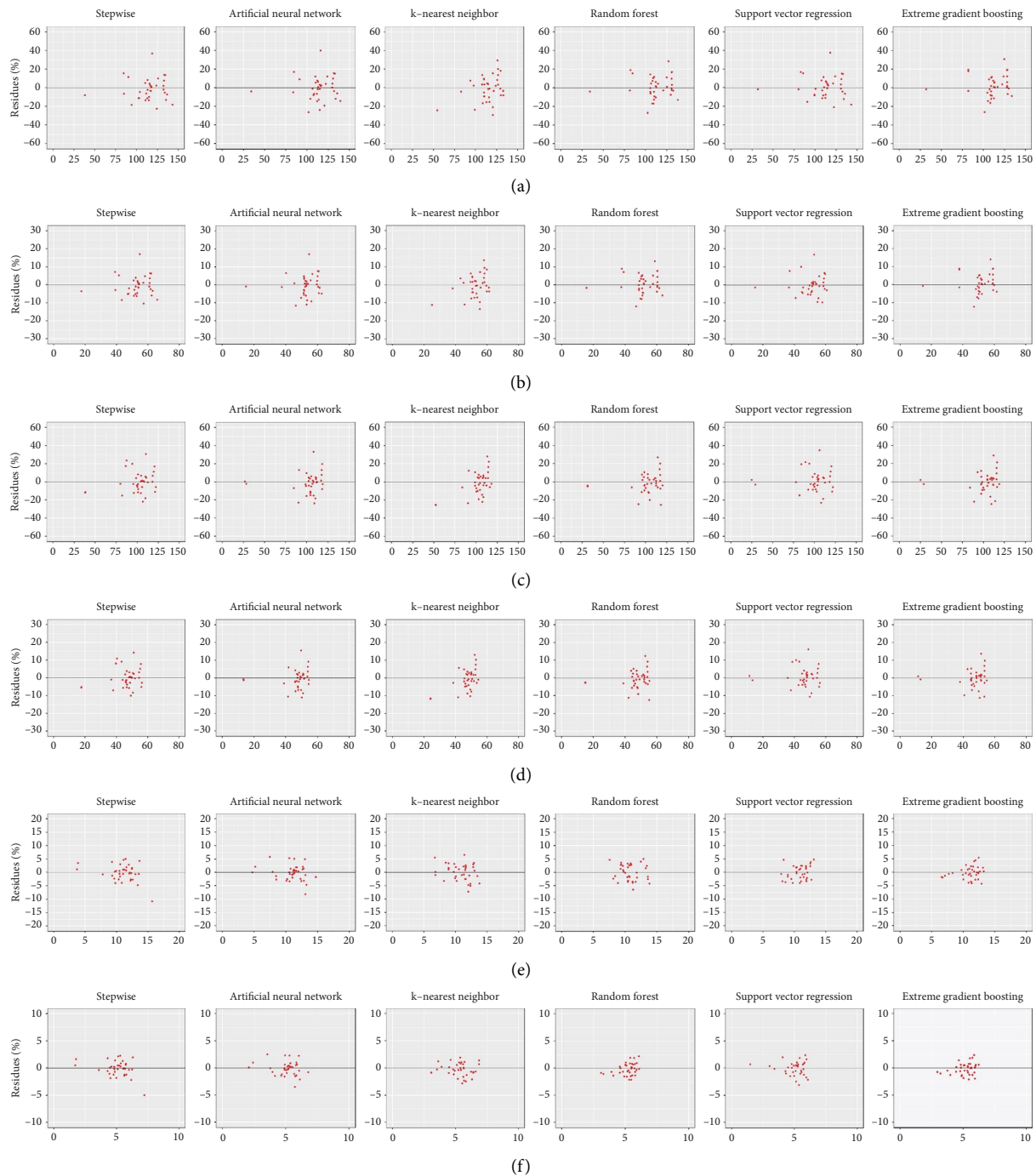


FIGURE 4: Predictive models of total aboveground biomass (a), aboveground carbon total (b), biomass trunk (c), carbon trunk (d), biomass canopy (e), and carbon canopy (f) by stepwise; artificial neural network, k-nearest neighbor, random forest, support vector regression, and extreme gradient boosting in *Eucalyptus* spp. in Itatinga, São Paulo, Brazil.

structural variables, precisely because it can capture non-linear relationships. [49] also reported high accuracy in estimating AGB of *Eucalyptus globulus* in Peru using LiDAR data collected with a DJI Zenmuse L1. The RF model achieved an R^2 of 0.76 on the training set and 0.66 on the test set. In contrast, multiple linear regression had an R^2 of 0.647, while elastic net reached an R^2 of 0.58 during training and 0.57 during testing [49]. However, in complex scenarios

involving large data volumes, algorithms such as XGB have outperformed RF in terms of precision and efficiency. A recent review on forest carbon stocks indicated that, although RF is widely used (around 88% of studies), XGB performed better in 75% of direct comparisons [51].

XGB's performance is attributed to its additive learning technique, which involves a learning process adjusted according to input data, followed by adaptation to residuals

TABLE 6: Agreement measures of the aboveground biomass and aboveground carbon (t·ha⁻¹) models by compartment in *Eucalyptus* spp.

Variable	Algorithm	Total basis			Fit					Validation			
		A	P	CC2	Λ	A	P	CC2	Λ	A	P	CC2	Λ
Total aboveground biomass	Stepwise	0.93	0.85	0.85	0.92	0.93	0.86	0.85	0.92	0.93	0.85	0.84	0.92
	ANN	0.95	0.89	0.89	0.94	0.96	0.91	0.91	0.95	0.91	0.82	0.81	0.90
	kNN	0.92	0.84	0.80	0.91	0.93	0.85	0.82	0.92	0.88	0.77	0.72	0.87
	RF	0.97	0.94	0.94	0.97	0.99	0.97	0.97	0.99	0.93	0.85	0.84	0.92
	SVR	0.93	0.86	0.86	0.92	0.93	0.86	0.85	0.92	0.93	0.86	0.86	0.93
	XGB	0.96	0.92	0.92	0.96	0.97	0.95	0.94	0.97	0.92	0.84	0.83	0.92
Total aboveground carbon	Stepwise	0.93	0.85	0.85	0.92	0.93	0.86	0.85	0.92	0.92	0.85	0.84	0.92
	ANN	0.94	0.88	0.87	0.94	0.95	0.90	0.89	0.95	0.91	0.82	0.80	0.90
	kNN	0.92	0.84	0.80	0.91	0.93	0.85	0.82	0.92	0.88	0.77	0.72	0.87
	RF	0.97	0.95	0.94	0.97	0.99	0.97	0.97	0.99	0.93	0.85	0.84	0.92
	SVR	0.85	0.71	0.71	0.83	0.93	0.86	0.85	0.92	0.59	0.18	0.18	0.31
	XGB	0.96	0.92	0.91	0.96	0.97	0.94	0.94	0.97	0.92	0.84	0.83	0.91
Trunk biomass	Stepwise	0.94	0.87	0.86	0.93	0.94	0.88	0.88	0.94	0.92	0.85	0.83	0.92
	ANN	0.95	0.91	0.90	0.95	0.96	0.93	0.92	0.96	0.93	0.86	0.85	0.93
	kNN	0.92	0.85	0.81	0.92	0.94	0.87	0.84	0.93	0.89	0.79	0.71	0.88
	RF	0.97	0.95	0.94	0.97	0.99	0.98	0.97	0.99	0.94	0.87	0.86	0.93
	SVR	0.94	0.87	0.87	0.93	0.94	0.88	0.88	0.94	0.93	0.86	0.85	0.93
	XGB	0.97	0.94	0.93	0.97	0.98	0.96	0.96	0.98	0.94	0.87	0.86	0.93
Trunk carbon	Stepwise	0.94	0.87	0.86	0.93	0.94	0.88	0.88	0.94	0.92	0.85	0.83	0.92
	ANN	0.95	0.90	0.90	0.95	0.96	0.92	0.92	0.96	0.93	0.86	0.85	0.92
	kNN	0.92	0.85	0.81	0.92	0.94	0.87	0.84	0.93	0.89	0.79	0.71	0.88
	RF	0.97	0.95	0.94	0.97	0.99	0.98	0.97	0.99	0.94	0.87	0.86	0.93
	SVR	0.94	0.87	0.87	0.93	0.94	0.88	0.88	0.94	0.93	0.86	0.85	0.93
	XGB	0.97	0.94	0.93	0.97	0.98	0.97	0.96	0.98	0.94	0.87	0.86	0.93
Canopy biomass	Stepwise	0.86	0.71	0.70	0.83	0.90	0.81	0.80	0.89	0.70	0.39	0.39	0.56
	ANN	0.87	0.74	0.72	0.85	0.91	0.83	0.81	0.90	0.72	0.44	0.42	0.61
	kNN	0.82	0.64	0.63	0.78	0.87	0.74	0.72	0.85	0.65	0.29	0.29	0.45
	RF	0.88	0.76	0.73	0.86	0.94	0.88	0.85	0.94	0.65	0.29	0.28	0.45
	SVR	0.76	0.51	0.48	0.68	0.76	0.53	0.51	0.69	0.73	0.46	0.38	0.63
	XGB	0.92	0.83	0.80	0.91	0.95	0.90	0.87	0.95	0.79	0.59	0.53	0.74
Canopy carbon	Stepwise	0.86	0.71	0.70	0.83	0.90	0.81	0.80	0.89	0.70	0.39	0.39	0.56
	ANN	0.87	0.75	0.73	0.85	0.91	0.83	0.81	0.91	0.72	0.44	0.42	0.61
	kNN	0.87	0.74	0.72	0.85	0.90	0.80	0.78	0.89	0.65	0.29	0.29	0.45
	RF	0.93	0.85	0.81	0.92	0.96	0.93	0.90	0.96	0.65	0.29	0.28	0.45
	SVR	0.84	0.68	0.66	0.81	0.87	0.73	0.72	0.85	0.73	0.46	0.38	0.63
	XGB	0.90	0.80	0.76	0.89	0.93	0.87	0.83	0.93	0.79	0.59	0.53	0.74

Note: A: coefficient of agreement; p: measure of agreement; CC2: improved concordance correlation coefficient; λ: agreement parameter. The bold values refer to the mean of each variable.

Abbreviations: ANN, artificial neural network; kNN, k-nearest neighbor; RF, random forest; SVR, support vector regression; XGB, extreme gradient boosting.

[52]. This adjustment process is repeated until stopping criteria are met. Thus, the algorithm is based on gradient-boosted decision trees, employing supervised ML with additive training strategies. This minimizes overfitting, optimizing the loss function not via first-order derivatives, but through an efficient second-order expression [52].

In forestry applications using UAV-LiDAR data, XGB can capture complex nonlinear relationships between height, point density, and intensity metrics, often outperforming RF in situations with high spatial heterogeneity or multilayered tree compartments [51]. However, its performance is strongly dependent on hyperparameter tuning (e.g., number of trees, maximum depth, learning rate) [52], requiring more computational resources than RF.

In the present study, XGB outperformed in the crown compartment (composed of leaves, live branches, and dead branches), whereas RF proved more efficient in the stem and

total compartments. XGB's ability to model complex nonlinear relationships may explain this result. These characteristics are present in the crown structure, where leaf and branch distribution creates variable height and density patterns captured by UAV-LiDAR metrics. Conversely, stem and total volume exhibit more linear relationships with UAV-LiDAR metrics, benefiting RF performance.

Studies on *Eucalyptus* plantations using UAV-LiDAR confirm these findings. For instance, [53] used XGB to estimate AGB of *Eucalyptus globoidea* and *Eucalyptus bosistoana*, employing structural crown metrics derived from LiDAR. The model achieved an R^2 of 0.903, outperforming RF and SVR. In previous studies using XGB, the combination of Landsat-8 and Sentinel-1A imagery as predictors for biomass estimation reached an R^2 of 0.75 and RMSE (%) of 38.28 in subtropical forests in China [54]. AGB in forests of Jharkhand, India, was estimated with an R^2 of

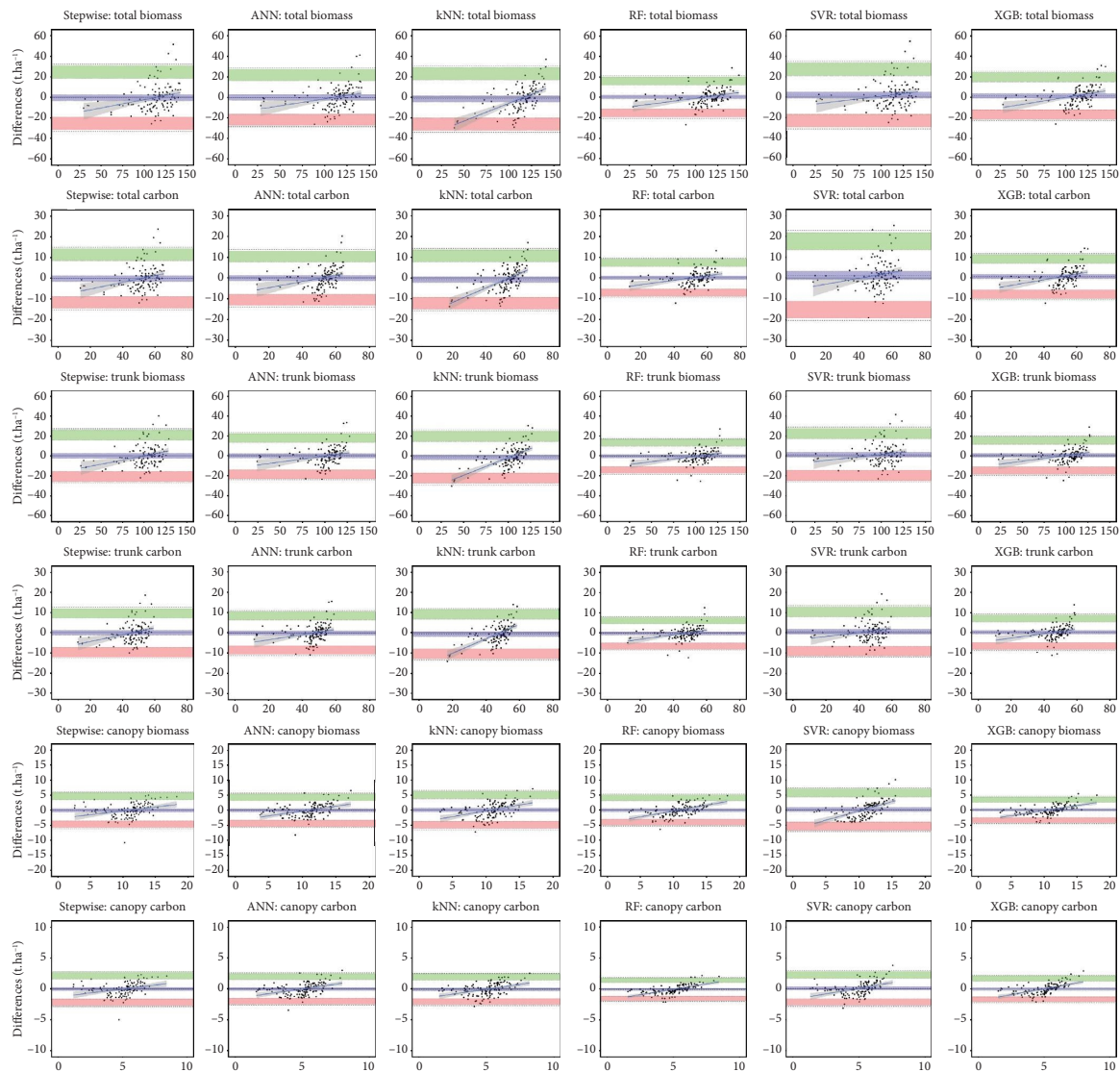


FIGURE 5: Bland-Altman plot for the predictive models of aboveground biomass and aboveground carbon in the total, trunk, and canopy compartments in *Eucalyptus* spp., generated by multiple linear regression and machine learning models. The colored bands represent the limits of agreement that encompass 95% of the data variation.

0.83 using vegetation indices derived from Sentinel imagery [55].

Overall, ANN was the third-best performing method. While it captured nonlinear relationships, the model showed high sensitivity to data quantity. Furthermore, [56] note that factors such as the size of test and validation sets (since ANNs require large datasets), distributed parallel processing, and adaptive nonlinear learning can affect ANN performance. Some of these characteristics make ANNs “black boxes,” complicating variable interpretation and function approximation convergence [56].

According to the literature, estimating AGB and carbon in *Eucalyptus* using UAV-LiDAR and ANNs yielded results similar to those of this study, as shown by [17]. In that study, AGB of *Eucalyptus globulus* was estimated with R^2 ranging from 0.85 to 0.92, depending on LiDAR point density and canopy characteristics.

However, despite efficiently modeling nonlinear patterns, ANNs are often outperformed by algorithms such as XGB in compartments with high spatial heterogeneity, like the crown. This is due to XGB's superior ability to optimize the loss function and adapt to complex residuals.

As with other algorithms, there is a gap in studies using ANNs and UAV-LiDAR metrics for predicting biomass and carbon in clonal *Eucalyptus* plantations in Brazil. Using other remote sensing data, [57] applied multispectral and SAR data to estimate biomass in *Eucalyptus urophylla* × *Eucalyptus grandis* in Minas Gerais, Brazil. Data from 206 inventory plots were used, and ANN-derived models achieved an R^2 of 0.95 and RMSE of 2.87%, with errors ranging from -8% to 4% [57].

Results for predicting biomass and carbon in the total and stem compartments of *Eucalyptus* genotypes indicate that, although linear regression outperformed some ML

methods (kNN and SVR), it relies on linear relationships between biomass or carbon and predictor variables, failing to capture the complex relationships present between biomass/carbon and UAV-LiDAR-derived variables. However, [49] note that multiple linear regression can provide a baseline for comparison with more complex methods, such as ML algorithms or nonlinear models.

In the present study, regression did not yield adequate estimates, unlike in other studies where it performed excellently in estimating biomass and carbon in *Eucalyptus* spp. In *Eucalyptus urophylla* × *Eucalyptus grandis* clones, models derived from this technique estimated carbon stock with adjusted R^2 above 0.98 and RMSE (%) below 12.86% [42]. In 6.5-year-old *Eucalyptus grandis*, simple linear regression models built using LiDAR metrics for carbon estimation achieved an adjusted R^2 of 0.97 [27].

Regarding algorithms with the highest errors, for SVR, the lack of a standardized approach to parameter optimization, as discussed by [58], may have resulted in inaccurate estimates. For kNN, optimization of the number of nearest neighbors, distance metrics, and weighting methods can improve performance [59], since the distance metric determines which training observations are closest to the target value based on predictor variables [60].

kNN showed consistent but lower performance compared to other algorithms, except SVR. This is due to its nonparametric nature and strong dependence on the choice of neighbors and distance metric. In forested regions, structural diversity directly affects neighbor proximity, which can increase variation in biomass estimates. Studies such as [54] indicate that kNN is sensitive to high-dimensional datasets and performs worse in the absence of significant spatial or structural correlation among neighboring plots.

In contrast, SVR exhibited superior fit under various conditions, corroborating the studies by [56, 61]. These studies highlighted SVR's ability to handle complex nonlinear relationships between UAV-LiDAR-derived variables and biomass. Using kernel functions, such as the radial basis function (RBF), allows SVR to detect nonlinear patterns more effectively than proximity-based methods like kNN. However, parameter calibration (C , ϵ , and γ) is essential and computationally expensive. Additionally, SVR is less intuitive and more restricted in interpretability compared to neighborhood-based methods.

Furthermore, some limitations of this study should be considered, such as the Zenmuse L1 sensor used. Sensor type, particularly pulse rate and return type, affects canopy penetration, point density, and flight altitude [21]. These factors may impact biomass estimation in very dense or heterogeneous crowns. Model performance may also be influenced by spatial heterogeneity of plots and limited data quantity.

Although conventional ML methods (RF, XGB, ANN, SVR, and kNN) were appropriate for this experimental study with a limited number of plots, recent literature indicates that modern deep learning (DL) techniques are more efficient for estimating AGB in complex scenarios with multimodal data. For instance, [62] applied adapted 3D convolutional neural networks (Minkowski CNNs) directly

to airborne LiDAR point clouds to estimate biomass and wood volume, observing considerable improvements over traditional statistical methods.

Similarly, [63] implemented ForestIQNet, a multimodal architecture integrating UAV multispectral images and 3D CHM models in a CNN-Transformer framework. This model achieved an R^2 above 0.89, compared to R^2 values of 0.78–0.82 obtained by RF and XGB. Reference [64] employed an attention-based DL model combined with GEDI, Sentinel-1, PALSAR-2, and Sentinel-2 data. This model outperformed RF in AGB estimation, with lower RMSE and improved spatial representation.

Therefore, it is recommended that future studies integrate additional LiDAR acquisition platforms, such as terrestrial laser scanning (TLS) and high-resolution UAV orthophotos. In this context, [17] showed that fusing airborne LiDAR and multispectral data allows for more accurate vegetation distribution, enhancing biomass estimation. Reference [22] proposed an automated framework for extracting individual tree attributes using high-density UAV-LiDAR, indicating that combining it with orthophotos increases inventory and monitoring accuracy. Complementarily, [65] used aerial digital camera and UAV-LiDAR fusion, demonstrating that integrating different data sources provides advantages for biomass estimation, suggesting potential for more detailed analyses across multiple tree compartments.

Finally, it should be emphasized that the use of the ABA approach applied in this research to *Eucalyptus* spp. plantations should consider the formation of strata (same species and planting date) and destructive sampling of AGB alongside seasonal UAV-LiDAR flights to obtain the metrics. It is recommended to follow a flight schedule, starting after planting and subsequently on an annual basis. With the proper application of these data in predictive models of AGB and carbon, the results generated can support decisions for monitoring the stock of these areas, as well as identifying their growth rate and role in addressing the climate emergency.

5. Conclusions

The RF algorithm was chosen from the modeling methods tested for quantifying AGB and aboveground carbon in different compartments (total, trunk, and canopy) of *Eucalyptus* spp. genotypes using UAV-LiDAR data to predict the total AGB (with mean of 110.97 t·ha⁻¹ at 4.5 years old) and total carbon (with mean of 51.16 t·ha⁻¹), as well as trunk biomass (with mean of 100.46 t·ha⁻¹) and trunk carbon (mean of 46.34 t·ha⁻¹). However, XGB was superior to the others for the canopy compartment, with mean of 10.72 t·ha⁻¹ of canopy biomass and 4.96 t·ha⁻¹ in mean of canopy carbon.

These results reinforce the effectiveness of UAV-LiDAR data associated with advanced modeling methods, such as ML algorithms, for quantifying AGB and carbon stocks in clonal *Eucalyptus* spp. plantations. The use of these tools can provide strategic information for sustainable management of forest resources. It is recommended to integrate this approach, considering the variability between genotypes, growth conditions, and plantation ages, in addition to

considering the seasonality of UAV-LiDAR data and different techniques for extracting UAV-LiDAR metrics, aiming to increase the accuracy of biomass and carbon estimates to assist in decision-making related to climate change mitigation.

Finally, for monitoring carbon stocks using UAV-LiDAR data, it is recommended to establish a flight schedule—initially postplanting and thereafter on an annual basis—integrated with targeted field sampling to enhance the robustness of ML models.

Data Availability Statement

Research data are not shared.

Conflicts of Interest

The authors declare no conflicts of interest.

Author Contributions

Conceptualization: Carla Talita Pertille, Alexandre Behling, Carlos Roberto Sanquetta, Otávio Camargo Campoe, Guerric Le Maire, Joannès Guillemot, Geovanni Malatesta Barros, Emerson Roberto Schoeninger, and Ana Paula Dalla Corte; methodology: Carla Talita Pertille, Alexandre Behling, Carlos Roberto Sanquetta, Otávio Camargo Campoe, Guerric Le Maire, Joannès Guillemot, Geovanni Malatesta Barros, Emerson Roberto Schoeninger, and Ana Paula Dalla Corte; software: Carla Talita Pertille, Alexandre Behling, and Ana Paula Dalla Corte; validation: Otávio Camargo Campoe, Guerric Le Maire, Joannès Guillemot, Geovanni Malatesta Barros, and Ana Paula Dalla Corte; formal analysis: Carla Talita Pertille, Alexandre Behling, and Ana Paula Dalla Corte; investigation: Carla Talita Pertille; resources: Otávio Camargo Campoe, Guerric Le Maire, Joannès Guillemot, Geovanni Malatesta Barros, Emerson Roberto Schoeninger; data curation: Carla Talita Pertille; writing—original draft preparation: Carla Talita Pertille, Alexandre Behling, Otávio Camargo Campoe, Guerric Le Maire, Joannès Guillemot, Geovanni Malatesta Barros, Emerson Roberto Schoeninger, and Ana Paula Dalla Corte; writing—review and editing: Carla Talita Pertille and Ana Paula Dalla Corte; visualization: Carla Talita Pertille; supervision: Ana Paula Dalla Corte; project administration: Otávio Camargo Campoe, Guerric Le Maire, Joannès Guillemot, Geovanni Malatesta Barros, Emerson Roberto Schoeninger.

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